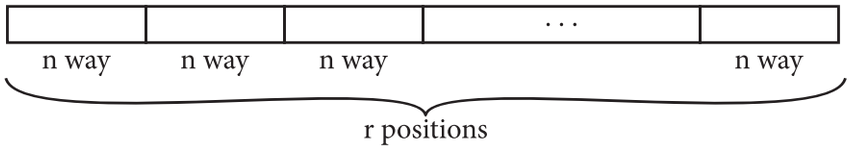
Proof. As in Theorem 4.1.



We can fill the first position with n objects. For the second position (still we can use the object used in first position), there are n objects, and so on the rth position can be filled with n objects.By the rule of product, The number of permutations of n different objects taken r at a time when repetition allowed is $n \times n \times n \times \cdots n$ (r times) $= n^r$. ■

4.4.2 Properties of Permutations

Property 1: ${}^nP_n = {}^n P_{n-1}$.

Proof. We have,

$$ {}^nP_{n-1} = \frac{n!}{(n-(n-1))!} = \frac{n!}{1!} = n! = \frac{n!}{(n-n)!} = {}^nP_n $$ ■

Property 2: ${}^nP_r = n \times {}^{n-1} P_{r-1}$

Proof. We have,

$$ n \times {}^{n-1}P_{r-1} = n \times \frac{(n-1)!}{((n-1)-(r-1))!} = \frac{n!}{(n-r)!} = {}^nP_r. $$

Note: Continuing the above process, we arrive at

$$ \begin{aligned} {}^nP_r &= n \times {}^{n-1}P_{r-1} = n \times (n-1) \times {}^{n-2} P_{r-2} \\ &= n \times (n-1) \times (n-2) \times {}^{n-3} P_{r-3} \times \cdots \times (n-(r-1))^{n-r} P_0 \\ &= n \times (n-1) \times (n-2) \times \cdots \times (n-r+1). \end{aligned} $$

$$ \boxed{{}^nP_r = n \times (n-1) \times (n-2) \times \cdots \times (n-r+1).} $$ ■

Property 3: ${}^nP_r = {}^{n-1}P_r + r \times {}^{n-1}P_{r-1}$

Proof. We have,

$$ \begin{aligned} {}^{n-1}P_r + r \times {}^{n-1}P_{r-1} &= \frac{(n-1)!}{((n-1)-r)!} + r\frac{(n-1)!}{(n-r)!} \\ &= \frac{(n-1)!}{(n-1-r)!} + r\frac{(n-1)!}{(n-r)!} \\ &= \frac{(n-1)! \times (n-r)}{(n-1-r)! \times (n-r)} + r\frac{(n-1)!}{(n-r)!} \\ &= \frac{(n-1)! \times (n-r)}{(n-r)!} + r\frac{(n-1)!}{(n-r)!} \\ &= \frac{(n-1)!\,((n-r)+r)}{(n-r)!} = \frac{(n-1)!n}{(n-r)!} \\ &= \frac{n!}{(n-r)!} = {}^nP_r \end{aligned} $$ ■